Lahore Urban Estates Analytics Warehouse

Star Schema Diagrams: conceptual, logical, physical, source ERD and architecture

DL3004 Data Warehousing and Business Intelligence · Final Project · Phase 1 · Version 2.1

FAST-NU Lahore · Fall 2026 · Prepared by: Junaid Minhas (23L-2559), Ahmad Waqar (23L-2634), Hanan Tiwana (23L-2536) and Eesha Khurram (23L-2641) · 16 October 2026

Contents

Figure 1	Conceptual model	page 2
Figure 2	Conceptual dimensional model (bus view)	page 3
Figure 3	Dimension hierarchies	page 4
Figure 4	Logical model · star 1 · fact_rent_invoice_line	page 5
Figure 5	Logical model · star 2 · fact_payment	page 6
Figure 6	Logical model · star 3 · fact_unit_occupancy_daily	page 7
Figure 7	Logical model · star 4 · fact_maintenance_request	page 8
Figure 8	Logical model · star 5 · fact_lease_event	page 9
Figure 9	Logical model · star 6 · fact_vacancy_incentive_coverage	page 10
Figure 10	Physical model · star 1 · fact_rent_invoice_line	page 11
Figure 11	Physical model · star 2 · fact_payment	page 12
Figure 12	Physical model · star 3 · fact_unit_occupancy_daily	page 13
Figure 13	Physical model · star 4 · fact_maintenance_request	page 14
Figure 14	Physical model · star 5 · fact_lease_event	page 15
Figure 15	Physical model · star 6 · fact_vacancy_incentive_coverage	page 16
Figure 16	Source ERD · src_pms (Lab 03 style)	page 17
Figure 17	End-to-end architecture	page 18

How to read the diagrams

- Crow's foot notation: || = exactly one, |o = zero or one, |{ = one or more, o{ = zero or more.
- Every star: each dimension connects directly to the fact; a dimension appears once per star and once per date role (the relationship label names the role, e.g. paid_date, due_date).
- PK = primary key (surrogate *_key in dimensions; *_sk or a composite key in facts). FK = foreign key to the dimension named in the note.
- Logical types: SK = integer surrogate key, money = PKR amount, flag = yes/no, text, date, integer, decimal.
- Annotations: 'measure: Additive / Semi-additive / Non-additive', 'hierarchy L1…L6', 'SCD2' and 'degenerate dimension'.
- Physical diagrams show PostgreSQL types, NULL / NOT NULL and the lineage columns source_system, batch_id, loaded_at.
- Special members: key -1 = Unknown, key -2 = Not applicable / not yet happened.
- Every diagram is generated from Mermaid source in diagrams/mermaid/*.mmd and can be edited and re-rendered (mermaid.live or the mmdc command line).

Figure 1. Conceptual model

Business entities and relationships only, no attributes. Payment to invoice line is many-to-many, which is why the logical model splits invoicing and payment into two facts.

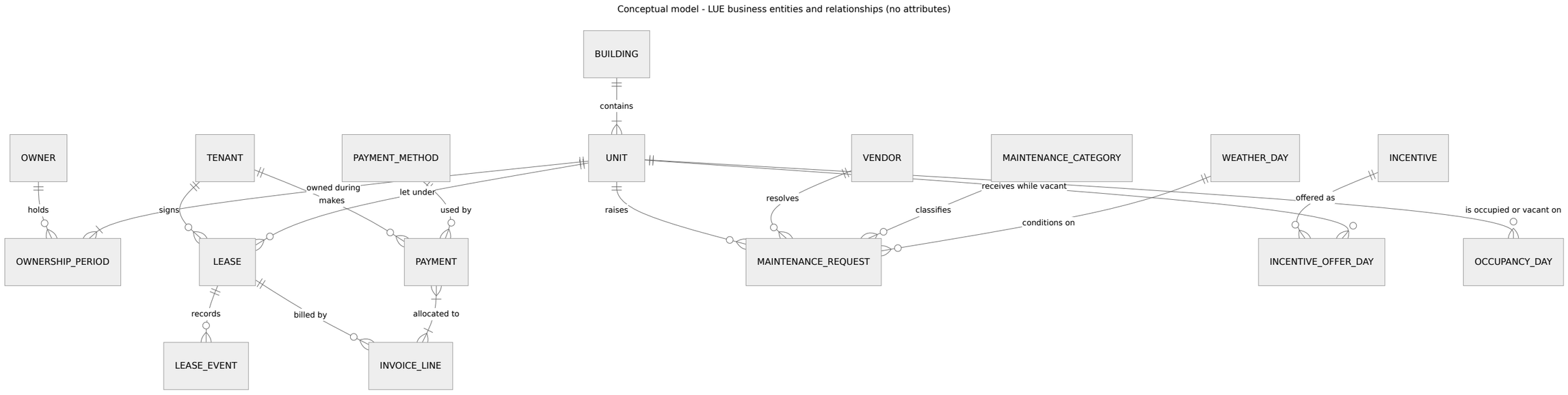


Figure 2. Conceptual dimensional model (bus view)

Five business processes, the six fact tables they produce and the 13 conformed dimensions each fact uses.

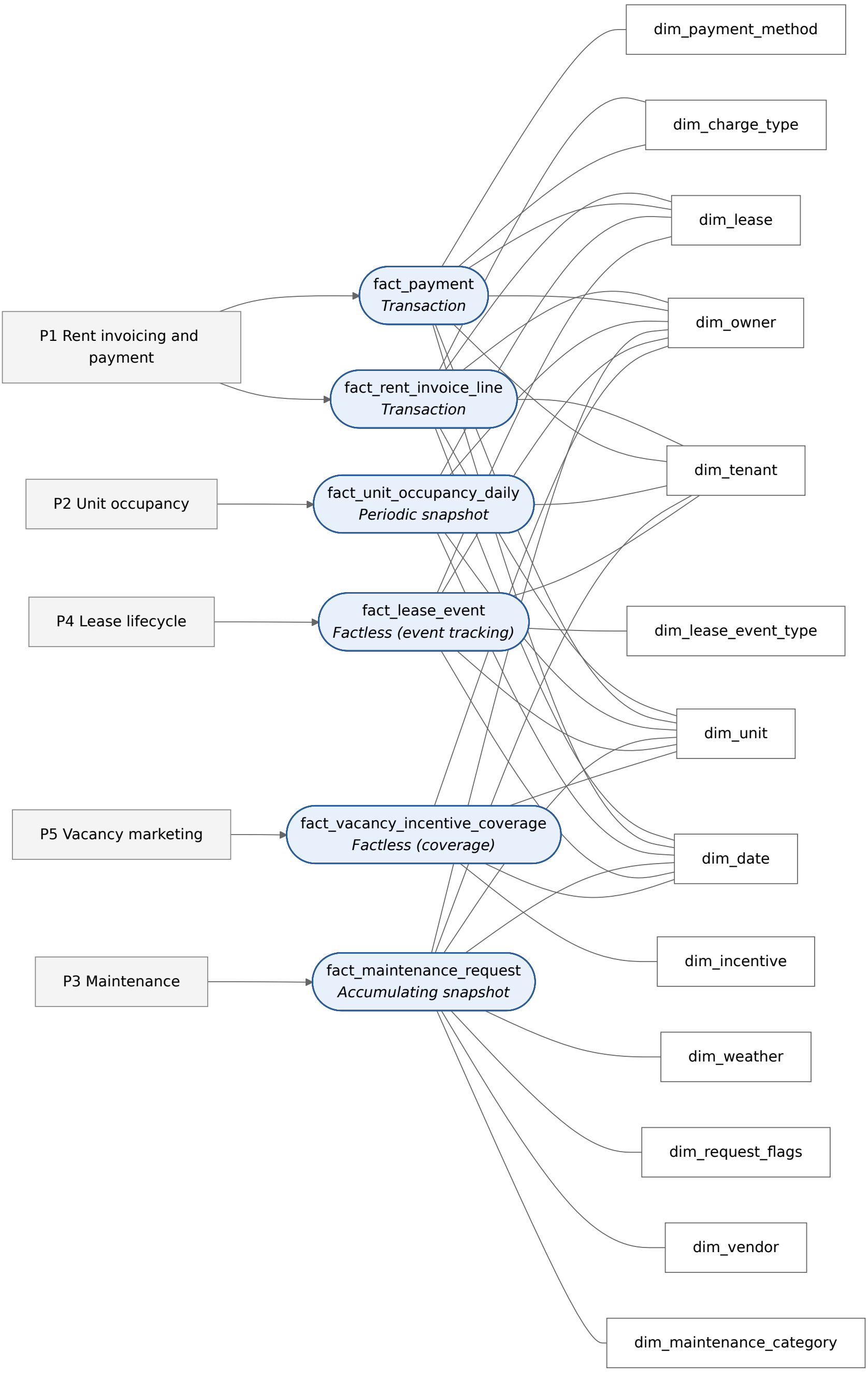


Figure 3. Dimension hierarchies

Drill-down paths from the lowest to the highest level. All are flattened into their dimension table (star schema, Lab 04 design rule).

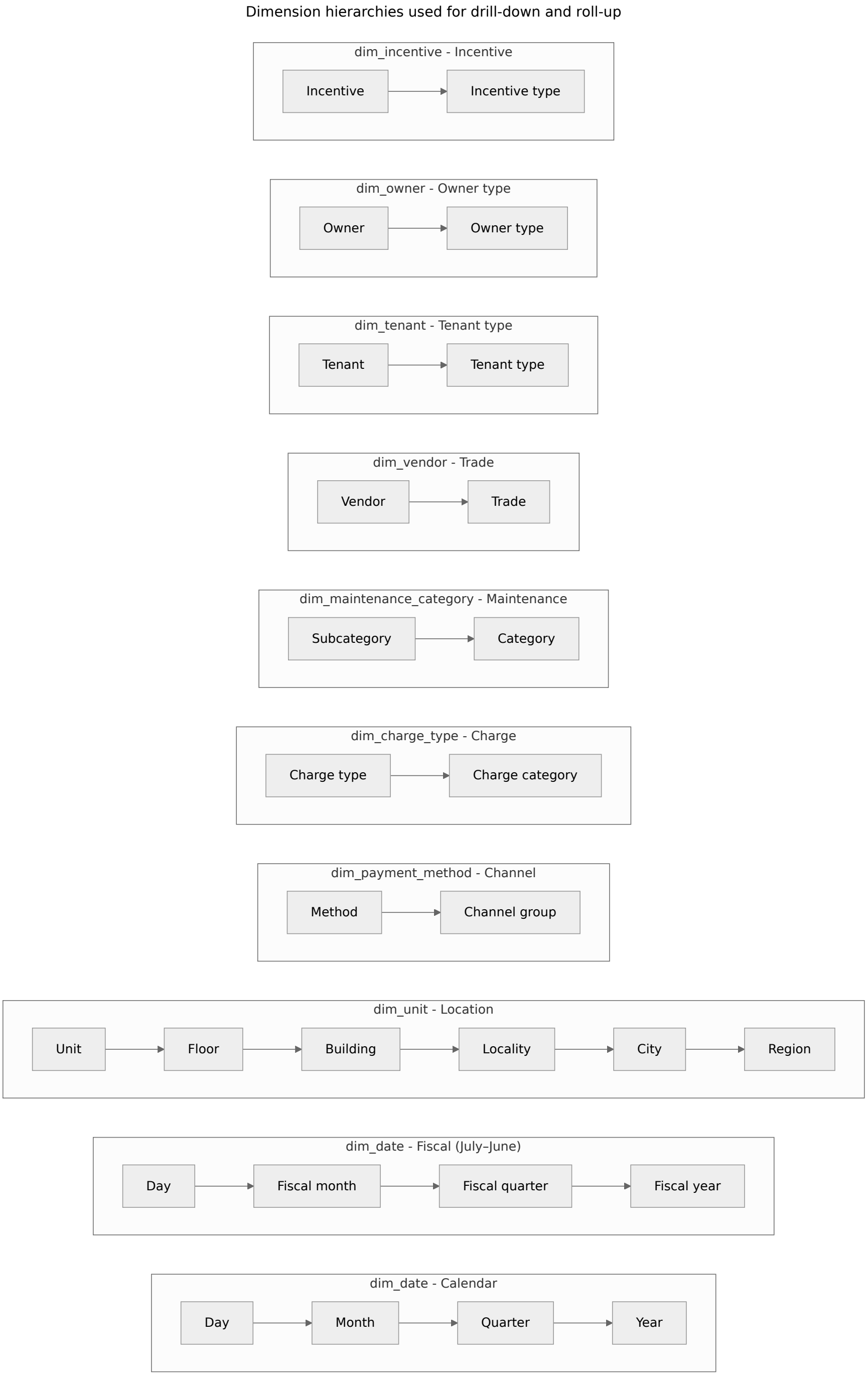


Figure 4. Logical model · star 1 · fact_rent_invoice_line

Transaction. Grain: One row represents one charge line (rent, service charge, late fee, utility or credit note) on one invoice issued to one lease. Logical types (SK, text, money, date, flag); PK/FK, measures with additivity, hierarchy levels, SCD Type 2 attributes and degenerate dimensions are annotated.

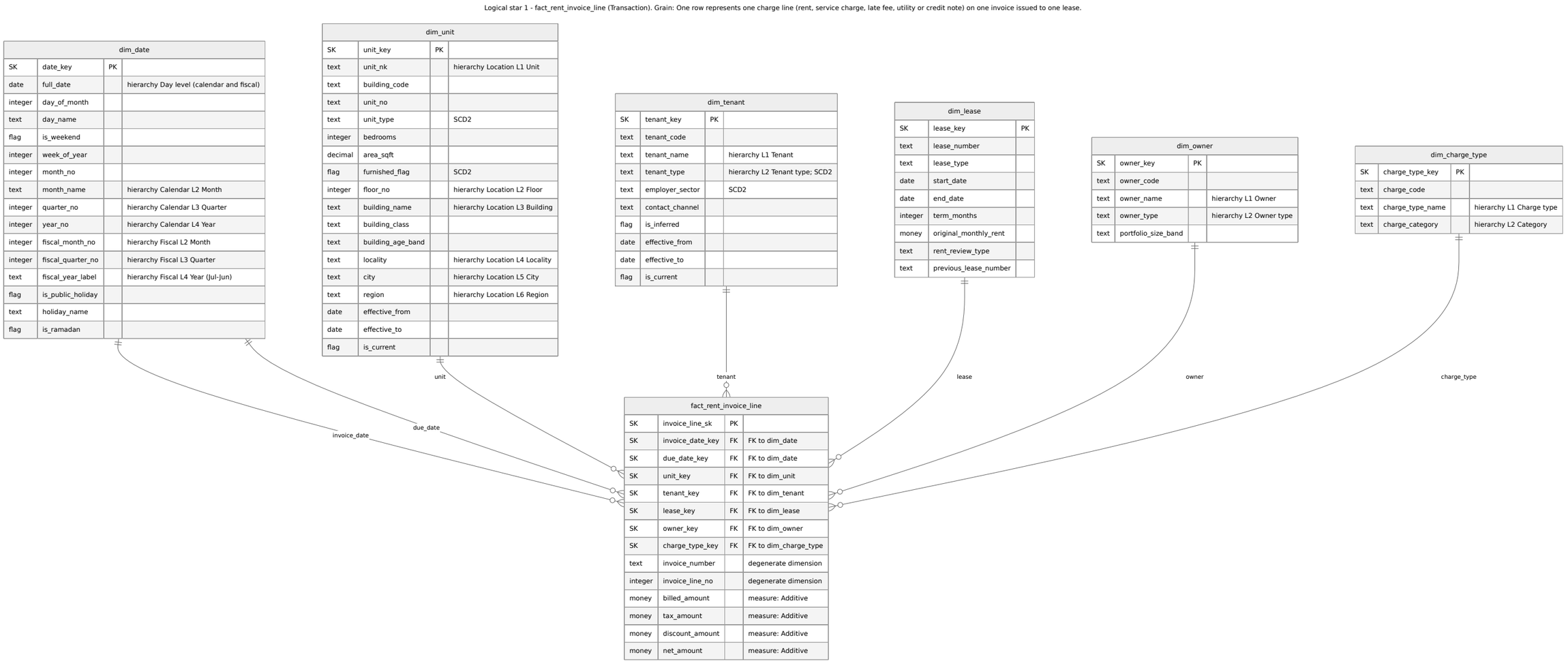


Figure 5. Logical model · star 2 · fact_payment

Transaction. Grain: One row represents one portion of one payment receipt allocated to one invoice line. Logical types (SK, text, money, date, flag); PK/FK, measures with additivity, hierarchy levels, SCD Type 2 attributes and degenerate dimensions are annotated.

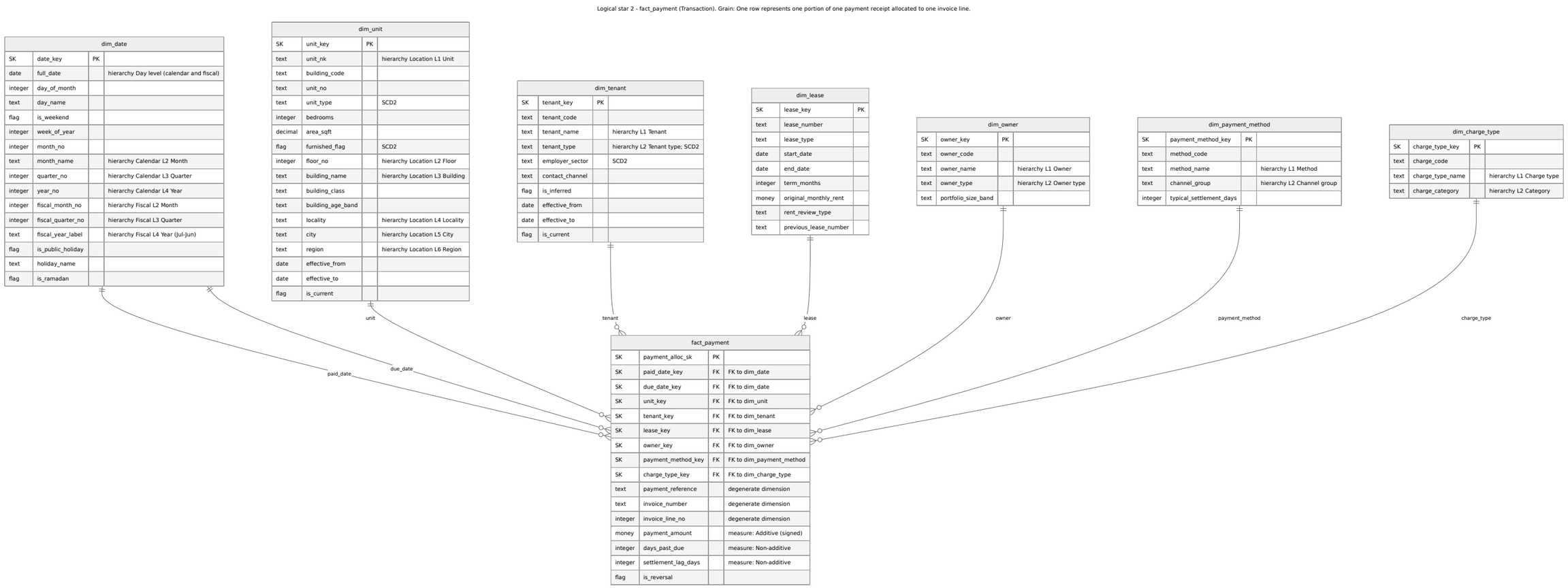


Figure 6. Logical model · star 3 · fact_unit_occupancy_daily

Periodic snapshot. Grain: One row represents one unit at the end of one calendar day, whether occupied or vacant. Logical types (SK, text, money, date, flag); PK/FK, measures with additivity, hierarchy levels, SCD Type 2 attributes and degenerate dimensions are annotated.

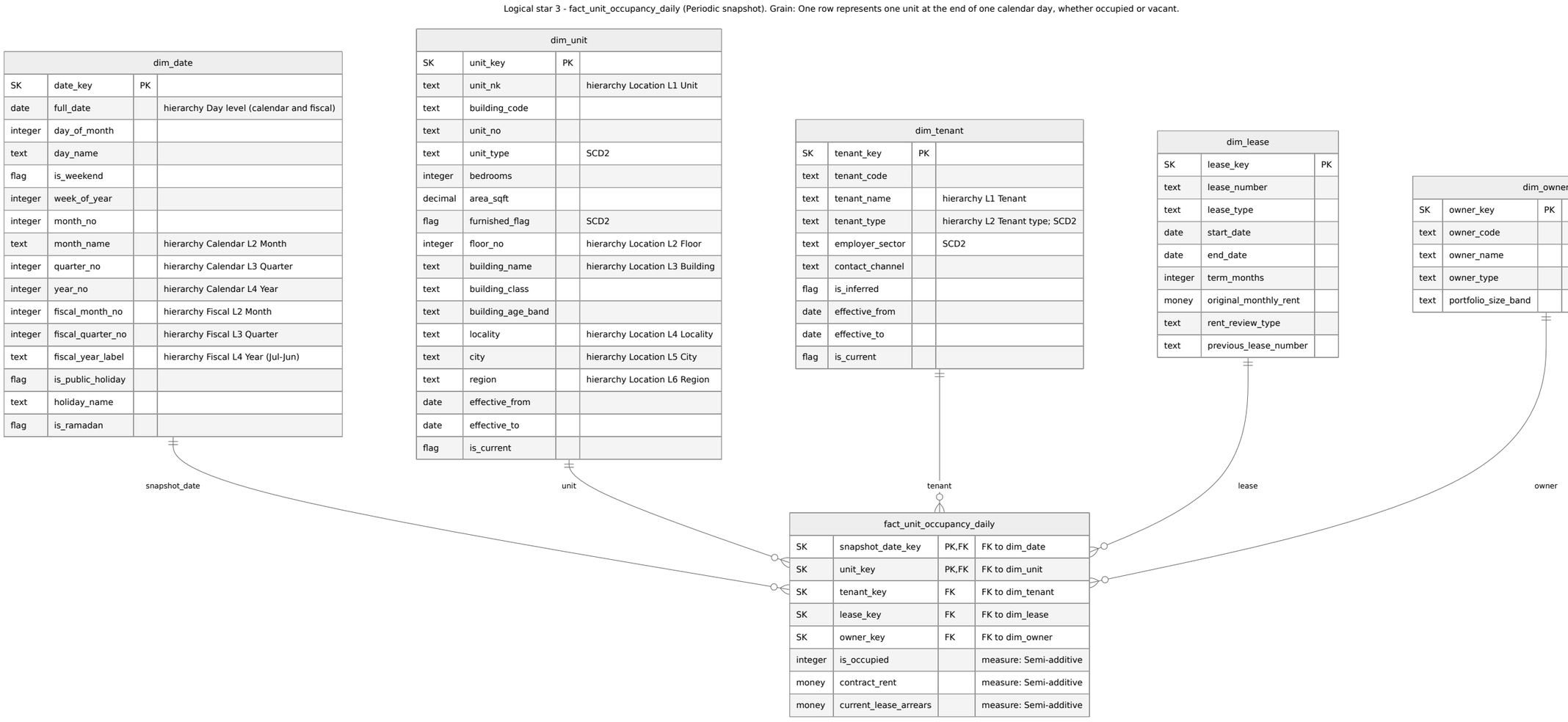


Figure 7. Logical model · star 4 · fact_maintenance_request

Accumulating snapshot. Grain: One row represents one maintenance request across its whole lifecycle, updated as it is assigned, resolved and verified. Logical types (SK, text, money, date, flag); PK/FK, measures with additivity, hierarchy levels, SCD Type 2 attributes and degenerate dimensions are annotated.

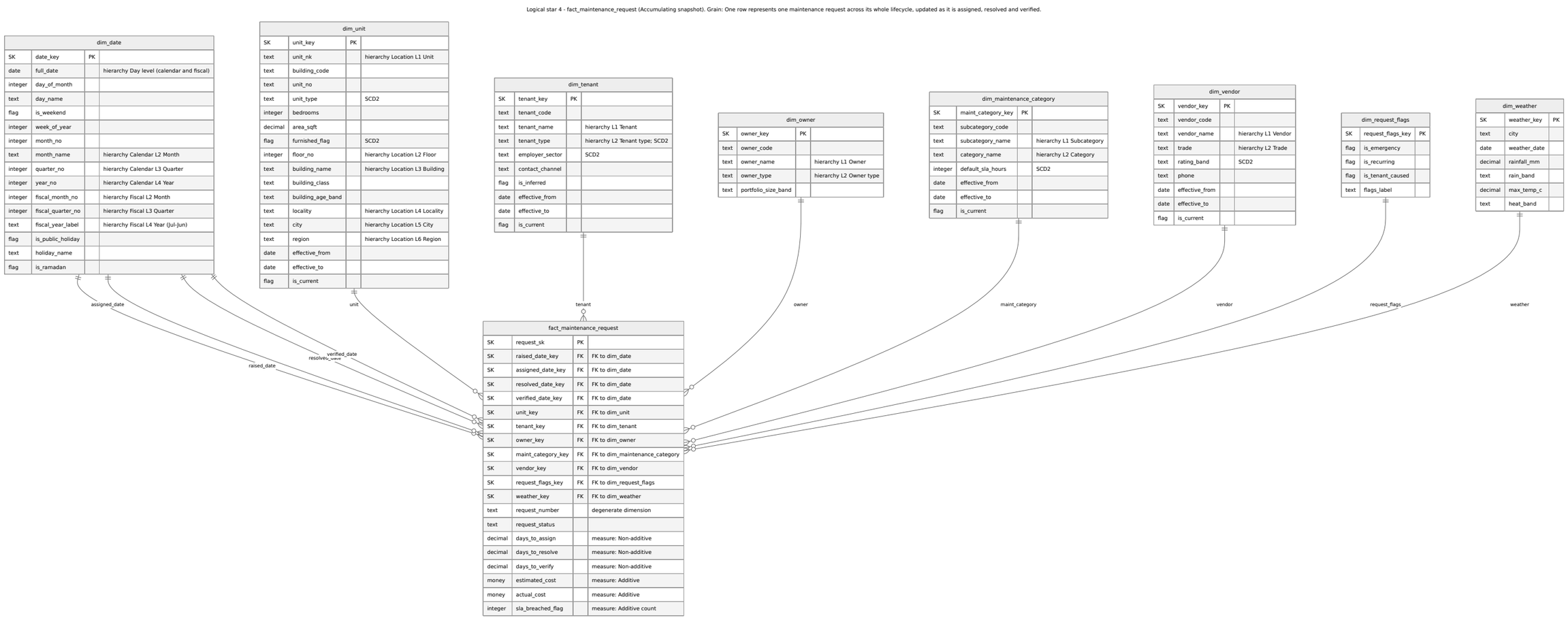


Figure 8. Logical model · star 5 · fact_lease_event

Factless (event tracking). Grain: One row represents one lifecycle event (signed, renewed, notice given or terminated) on one lease. Logical types (SK, text, money, date, flag); PK/FK, measures with additivity, hierarchy levels, SCD Type 2 attributes and degenerate dimensions are annotated.

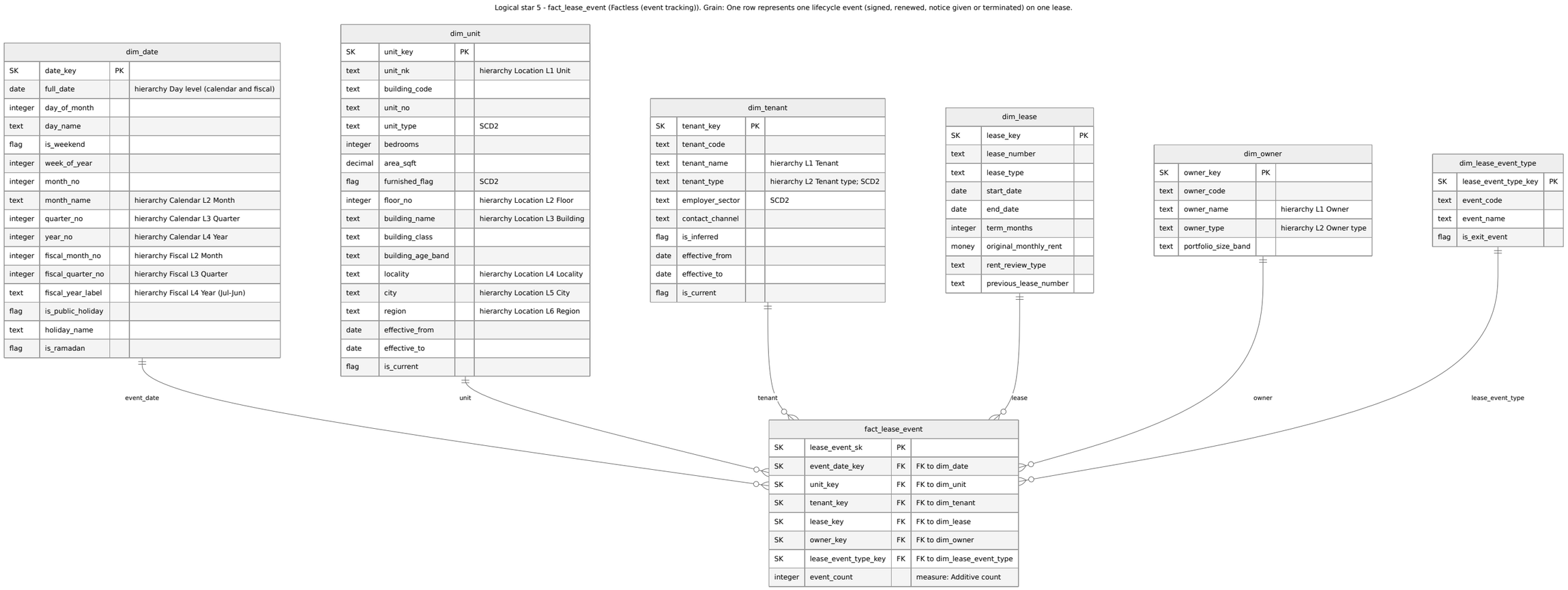


Figure 9. Logical model · star 6 · fact_vacancy_incentive_coverage

Factless (coverage). Grain: One row represents one incentive offer being available on one vacant unit on one calendar day. Logical types (SK, text, money, date, flag); PK/FK, measures with additivity, hierarchy levels, SCD Type 2 attributes and degenerate dimensions are annotated.

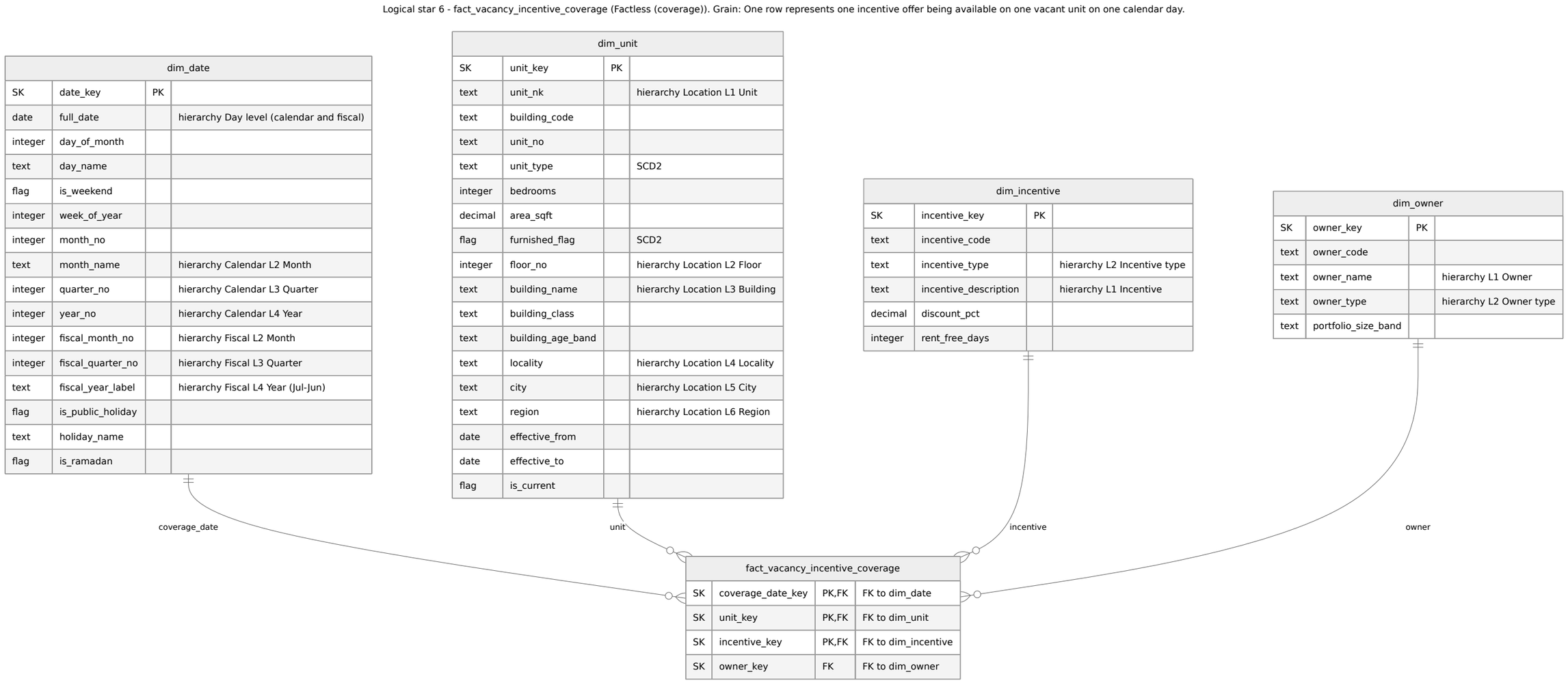


Figure 10. Physical model · star 1 · fact_rent_invoice_line

PostgreSQL 16 data types, primary and foreign keys, nullability and lineage columns, exactly as in sql/physical_model.sql. Grain: One row represents one charge line (rent, service charge, late fee, utility or credit note) on one invoice issued to one lease.

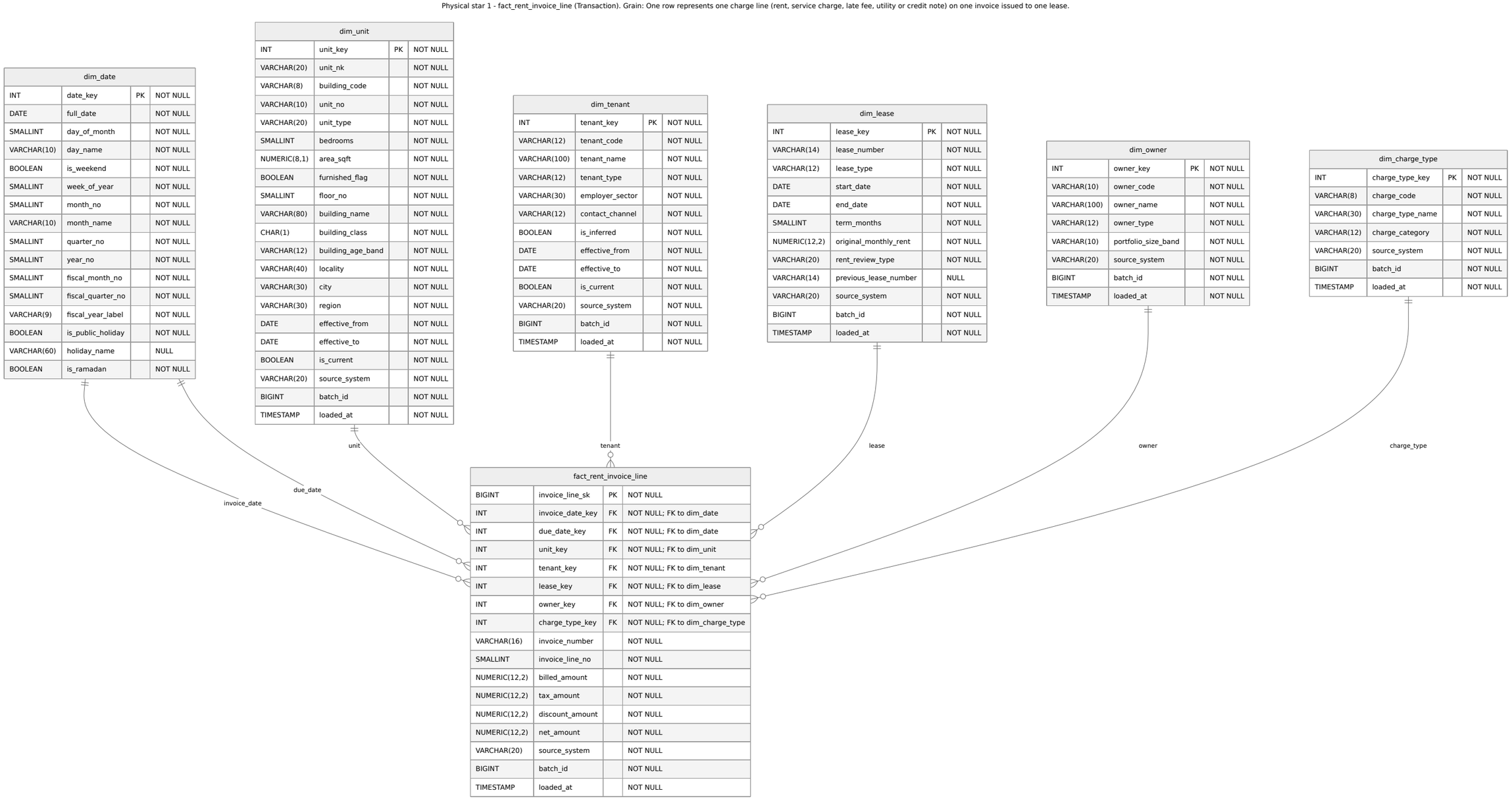


Figure 11. Physical model · star 2 · fact_payment

PostgreSQL 16 data types, primary and foreign keys, nullability and lineage columns, exactly as in sql/physical_model.sql. Grain: One row represents one portion of one payment receipt allocated to one invoice line.

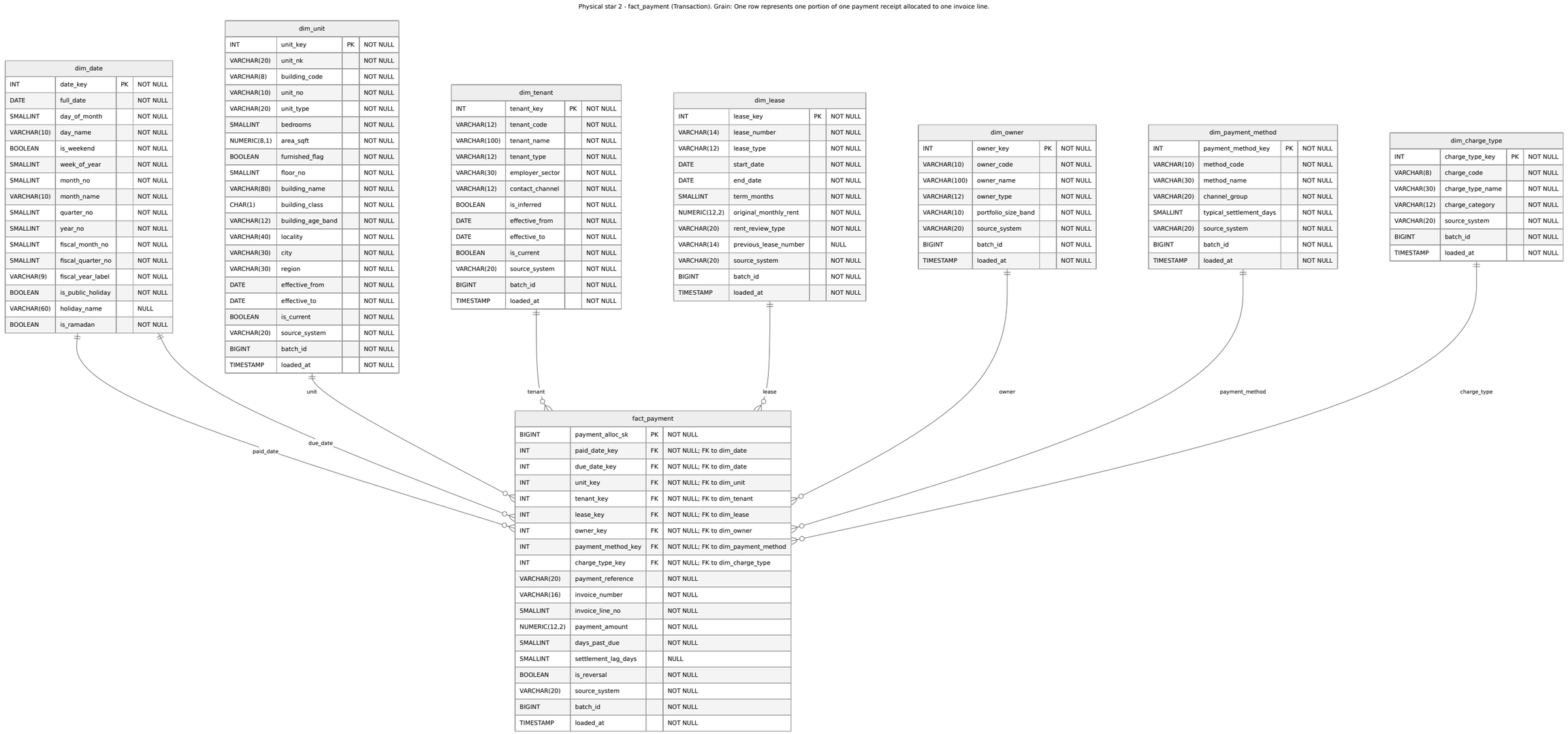


Figure 12. Physical model · star 3 · fact_unit_occupancy_daily

PostgreSQL 16 data types, primary and foreign keys, nullability and lineage columns, exactly as in sql/physical_model.sql. Grain: One row represents one unit at the end of one calendar day, whether occupied or vacant.

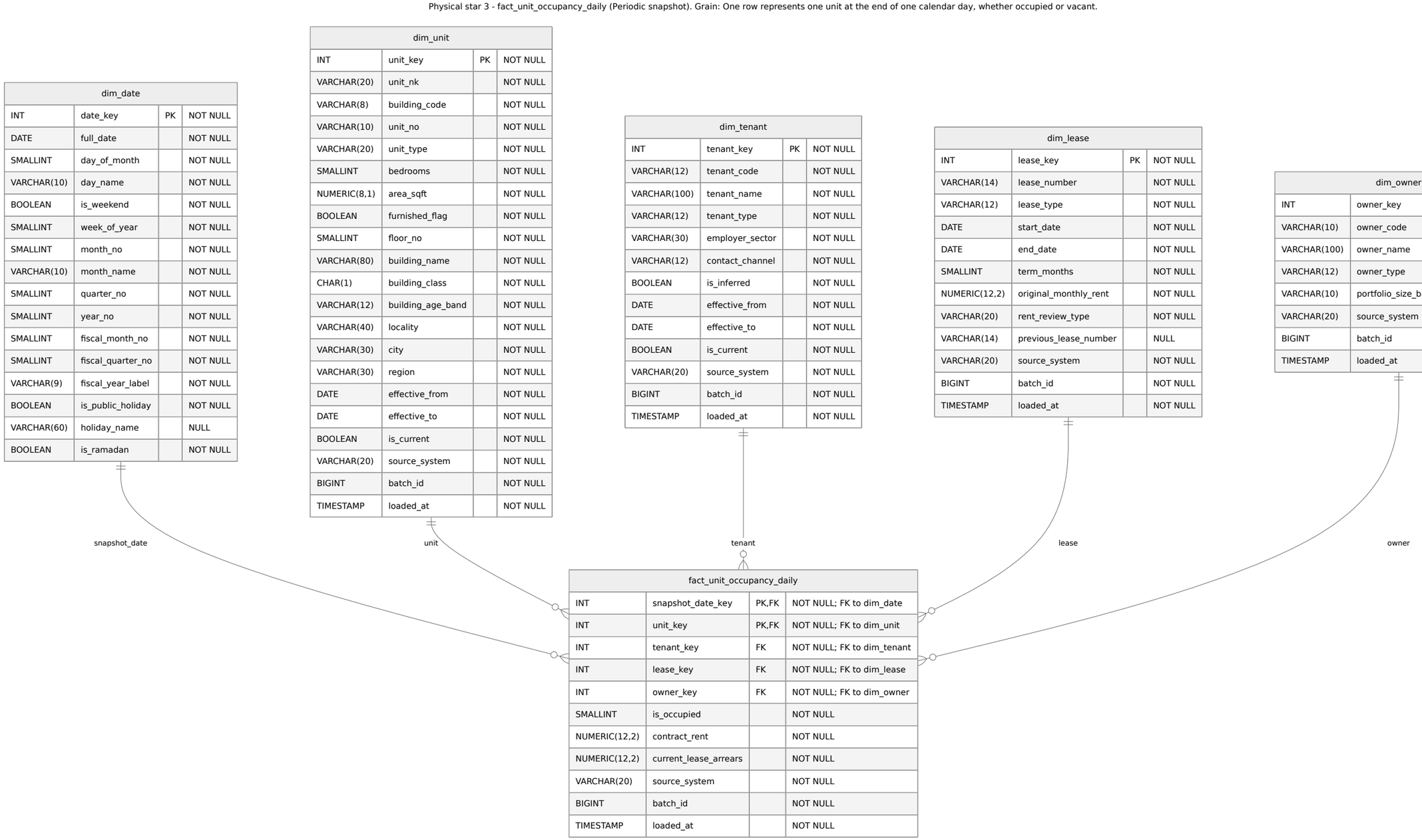


Figure 13. Physical model · star 4 · fact_maintenance_request

PostgreSQL 16 data types, primary and foreign keys, nullability and lineage columns, exactly as in sql/physical_model.sql. Grain: One row represents one maintenance request across its whole lifecycle, updated as it is assigned, resolved and verified.

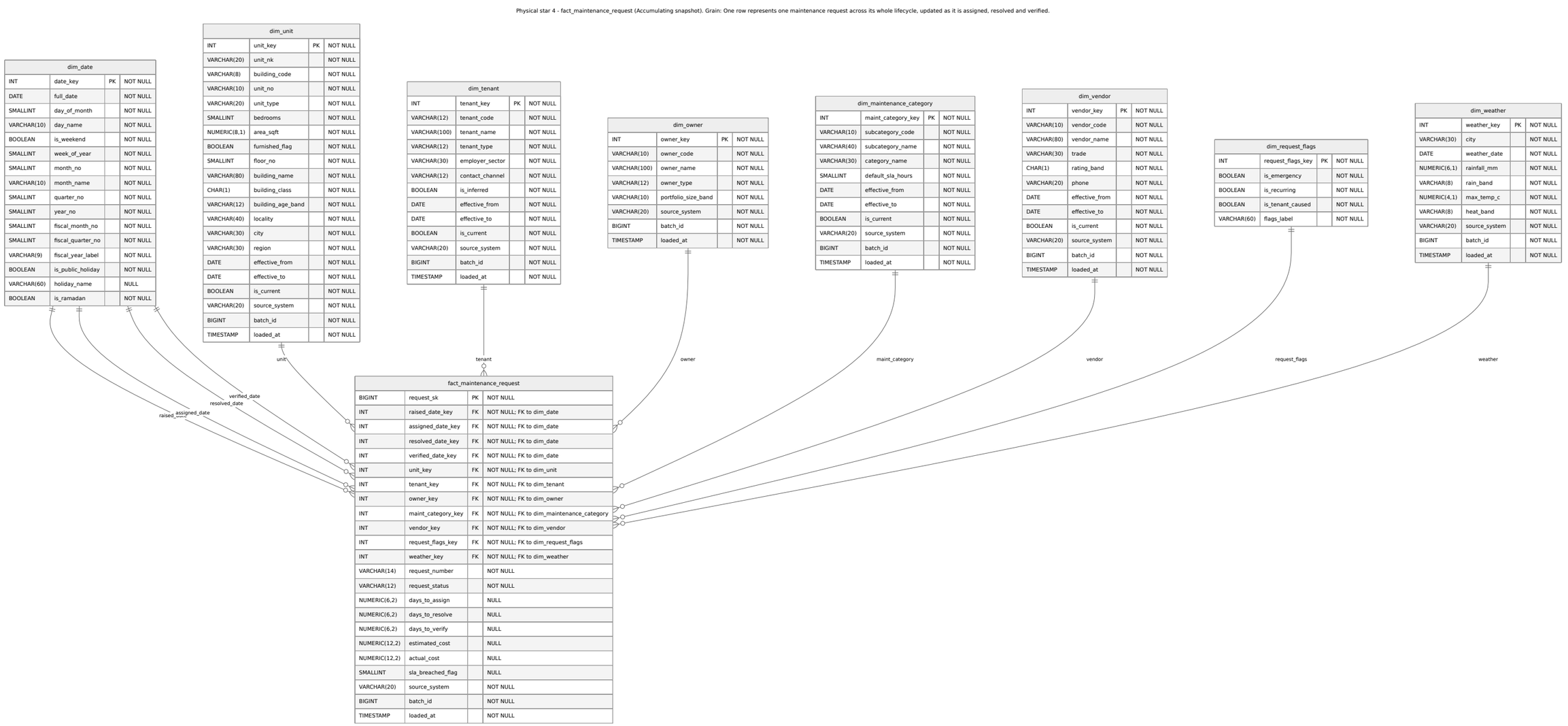


Figure 14. Physical model · star 5 · fact_lease_event

PostgreSQL 16 data types, primary and foreign keys, nullability and lineage columns, exactly as in sql/physical_model.sql. Grain: One row represents one lifecycle event (signed, renewed, notice given or terminated) on one lease.

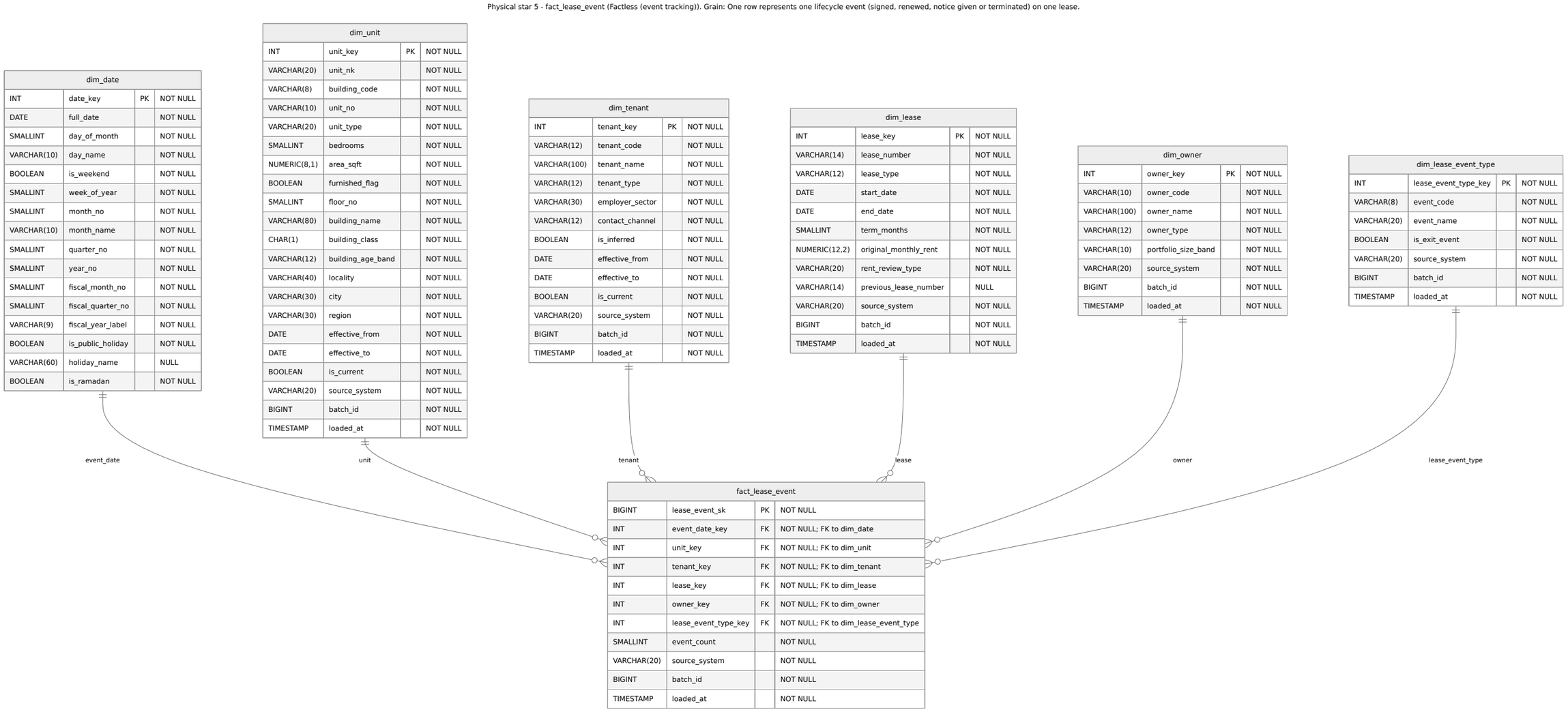


Figure 15. Physical model · star 6 · fact_vacancy_incentive_coverage

PostgreSQL 16 data types, primary and foreign keys, nullability and lineage columns, exactly as in sql/physical_model.sql. Grain: One row represents one incentive offer being available on one vacant unit on one calendar day.

Physical star 6 - fact_vacancy_incentive_coverage (Factless (coverage)). Grain: One row represents one incentive offer being available on one vacant unit on one calendar day.

dim_date			
INT	date_key	PK	NOT NULL
DATE	full_date		NOT NULL
SMALLINT	day_of_month		NOT NULL
VARCHAR(10)	day_name		NOT NULL
BOOLEAN	is_weekend		NOT NULL
SMALLINT	week_of_year		NOT NULL
SMALLINT	month_no		NOT NULL
VARCHAR(10)	month_name		NOT NULL
SMALLINT	quarter_no		NOT NULL
SMALLINT	year_no		NOT NULL
SMALLINT	fiscal_month_no		NOT NULL
SMALLINT	fiscal_quarter_no		NOT NULL
VARCHAR(9)	fiscal_year_label		NOT NULL
BOOLEAN	is_public_holiday		NOT NULL
VARCHAR(60)	holiday_name		NULL
BOOLEAN	is_ramadan		NOT NULL

dim_unit			
INT	unit_key	PK	NOT NULL
VARCHAR(20)	unit_nk		NOT NULL
VARCHAR(8)	building_code		NOT NULL
VARCHAR(10)	unit_no		NOT NULL
VARCHAR(20)	unit_type		NOT NULL
SMALLINT	bedrooms		NOT NULL
NUMERIC(8,1)	area_sqft		NOT NULL
BOOLEAN	furnished_flag		NOT NULL
SMALLINT	floor_no		NOT NULL
VARCHAR(80)	building_name		NOT NULL
CHAR(1)	building_class		NOT NULL
VARCHAR(12)	building_age_band		NOT NULL
VARCHAR(40)	locality		NOT NULL
VARCHAR(30)	city		NOT NULL
VARCHAR(30)	region		NOT NULL
DATE	effective_from		NOT NULL
DATE	effective_to		NOT NULL
BOOLEAN	is_current		NOT NULL
VARCHAR(20)	source_system		NOT NULL
BIGINT	batch_id		NOT NULL
TIMESTAMP	loaded_at		NOT NULL

dim_incentive			
INT	incentive_key	PK	NOT NULL
VARCHAR(10)	incentive_code		NOT NULL
VARCHAR(20)	incentive_type		NOT NULL
VARCHAR(80)	incentive_description		NOT NULL
NUMERIC(5,2)	discount_pct		NOT NULL
SMALLINT	rent_free_days		NOT NULL
VARCHAR(20)	source_system		NOT NULL
BIGINT	batch_id		NOT NULL
TIMESTAMP	loaded_at		NOT NULL

dim_owner			
INT	owner_key	PK	NOT NULL
VARCHAR(10)	owner_code		NOT NULL
VARCHAR(100)	owner_name		NOT NULL
VARCHAR(12)	owner_type		NOT NULL
VARCHAR(10)	portfolio_size_band		NOT NULL
VARCHAR(20)	source_system		NOT NULL
BIGINT	batch_id		NOT NULL
TIMESTAMP	loaded_at		NOT NULL

coverage_date

unit

incentive

owner

fact_vacancy_incentive_coverage			
INT	coverage_date_key	PK,FK	NOT NULL; FK to dim_date
INT	unit_key	PK,FK	NOT NULL; FK to dim_unit
INT	incentive_key	PK,FK	NOT NULL; FK to dim_incentive
INT	owner_key	FK	NOT NULL; FK to dim_owner
VARCHAR(20)	source_system		NOT NULL
BIGINT	batch_id		NOT NULL
TIMESTAMP	loaded_at		NOT NULL

Figure 16. Source ERD · src_pms (Lab 03 style)

The normalised operational schema of the PropertyHub source system, with primary keys, foreign keys and cardinalities. Shown to make clear that the warehouse is designed around business processes, not copied from these tables.

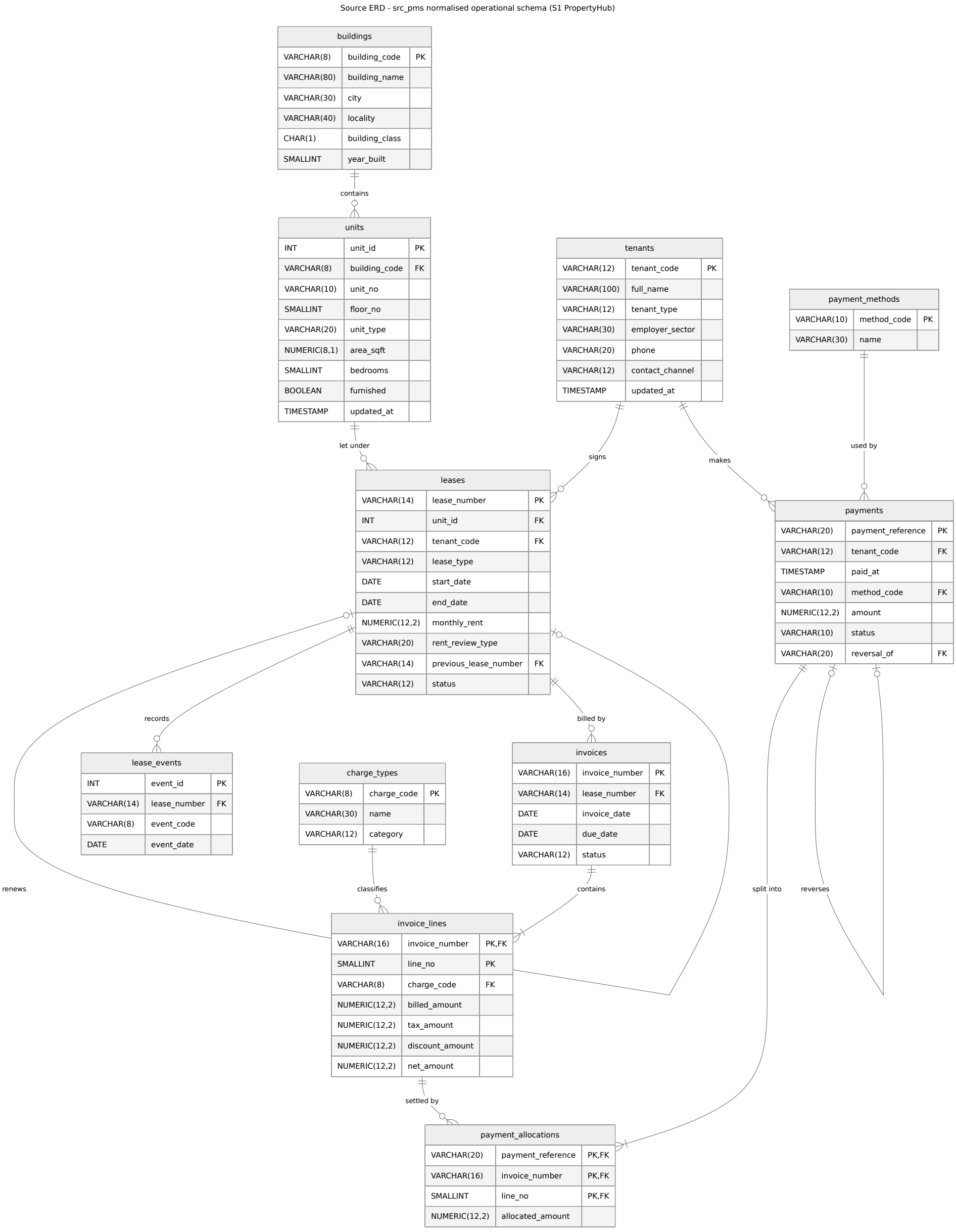


Figure 17. End-to-end architecture

The pipeline mapped to the 11 sections of the reference architecture diagram: sources, orchestration, Bronze, quality checks and quarantine, Silver, SCD processing, warehouse, performance, Gold marts, analytics and BI, audit, and lineage.

